

REVIEW QUESTIONS

3.1 State whether the following statements are *true* or *false*.

- (a) All variables must be given a type when they are declared. [LO 3.2 E]
- (b) ANSI C treats the variables **name** and **Name** to be same. [LO 3.2 E]
- (c) Character constants are coded using double quotes. [LO 3.2 E]
- (d) The keyword **void** is a data type in C. [LO 3.3 E]
- (e) Character constants are enclosed in single quotes while string constants are enclosed in double quotes. [LO 3.2 M]
- (f) Initialization is the process of assigning a value to a variable at the time of declaration. [LO 3.4 E]
- (g) The **scanf** function can be used to read only one value at a time. [LO 3.4 E]
- (h) C allows its keywords to be also used as identifiers. [LO 3.1 M]
- (i) Auto variables are by default initialized to 0 as soon as they are declared. [LO 3.4 H]
- (j) Floating point constants, by default, denote **float** type values. [LO 3.2 M]
- (k) Like variables, constants have a type. [LO 3.2 M]
- (l) All **static** variables are automatically initialized to zero. [LO 3.4 M]

3.2 Fill in the blanks with appropriate words.

- (a) A variable can be made constant by declaring it with the qualifier _____ at the time of initialization. [LO 3.5 E]
- (b) _____ is the largest value that an unsigned short int type variable can store. [LO 3.3 M]
- (c) A global variable is also known as _____ variable. [LO 3.4 M]
- (d) The keyword _____ can be used to create a data type identifier. [LO 3.4 H]
- (e) If the number 987612347 is to be used as an unsigned long integer then it must be appended by _____. [LO 3.3 E]
- (f) _____ and _____ are used to define user-defined data types in C. [LO 3.3 M]
- (g) The largest positive integer value can be stored in _____ type variable. [LO 3.3 H]

3.3 What are trigraph characters? How are they useful? [LO 3.1 E]

3.4 Describe the four basic data types. How could we extend the range of values they represent? [LO 3.3 E]

3.5 What is an unsigned integer constant? What is the significance of declaring a constant unsigned? [LO 3.4 M]

3.6 Describe the characteristics and purpose of escape sequence characters. [LO 3.2 M]

3.7 What is a variable and what is meant by the "value" of a variable? [LO 3.2 E]

3.8 How do variables and symbolic names differ? [LO 3.5 E]

3.9 State the differences between the declaration of a variable and the definition of a symbolic name. [LO 3.5 M]

3.10 What are the qualifiers that an **int** can have at a time? [LO 3.3 E]

3.11 A programmer would like to use the word DPR to declare all the double-precision floating point values in his program. How could he achieve this? [LO 3.4 M]

3.12 What are enumeration variables? How are they declared? What is the advantage of using them in a program? [LO 3.4 E]

3.13 Describe the purpose of the qualifiers **const** and **volatile**. [LO 3.5 M]

3.14 When dealing with very small or very large numbers, what steps would you take to improve the accuracy of the calculations? [LO 3.4 H]

3.15 Which of the following are invalid constants and why? [LO 3.2 M]

- | | | |
|--------|----------------|------------|
| 0.0001 | 5×1.5 | 99999 |
| +100 | 75.45 E-2 | "15.75" |
| -45.6 | -1.79 e + 4 | 0.00001234 |

3.16 Which of the following are invalid variable names and why? [LO 3.2 M]

- | | | | |
|---------|------------|-----------|--------------|
| Minimum | First.name | n1+n2 | &name |
| doubles | 3rd_row | n\$ | Row1 |
| float | Sum Total | Row Total | Column-total |

3.17 What would be the value of x after execution of the following statements? [LO 3.4 H]

```
int x, y = 10;
char z = 'a';
x = y + z;
```

3.18 Explain the following with examples:

(a) Enumerated types

(b) Type def

[LO 3.4 E]

[LO 3.4 E]

3.19 Distinguish between the following:

- (a) Global and local variables
- (b) Initialization and assignment of variables
- (c) Automated and static variables

[LO 3.4 E]

[LO 3.4 E]

[LO 3.4 M]